

Hybrid Qubit–Cavity Universal Control in Dispersive cQED: Follow-Up Optimization, Waveform Realization, and Deployability Study

cQED Autonomous Research Platform
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Abstract

This report is a follow-up to the consolidated hybrid qubit–cavity control study. The original question was which gate family looks best on the logical block $\{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle, |e, 1\rangle\}$. The present question is stricter: what are the actual optimized solutions, what pulse waveforms do the structured gates imply, and how much evidence do we have that the reported optima are meaningful rather than shallow local minima? We study a dispersive transmon–storage cavity system with $\omega_q/2\pi = 6.150$ GHz, $\omega_c/2\pi = 5.241$ GHz, $\chi/2\pi = -2.84$ MHz, $\chi'/2\pi = -21$ kHz, $K/2\pi = -28$ kHz, and $\alpha/2\pi = -255$ MHz using `cqed_sim` for all decomposition, pulse-construction, and optimal-control steps.

The main new result is that the previous structured U_{target} plateau was not the final story. Adding one more displacement gate to the best depth-11 $D+R_q$ +SQR ansatz lifts the logical process fidelity from 0.9212 to 0.9953 while reducing average leakage from 0.1050 to 0.0065. Repeated multistarts collapse onto the same optimum, strongly suggesting that the depth-12 structured SQR solution is near-saturated within the tested ansatz family. By contrast, the corresponding depth-12 $D + R_q$ + ConditionalPhaseSQR family only reaches 0.9135, so the extra displacement helps but does not close the expressivity gap. On the pulse-backed side, a tuned logical-window SNAP reaches closed-system average state fidelity 0.9857 and noisy average state fidelity 0.9694, whereas the constructive “selective π plus broadband π ” shortcut fails badly at ≈ 0.60 . The native χ -wait entangler remains essentially exact at 0.9999999 in 256 ns. The experiment-facing conclusion is now sharper: native χ -wait remains the deployable entangler, selective SNAP remains the cleanest local logical primitive, and the newly identified depth-12 SQR circuit is the most credible structured universal-control candidate worth pulse-level implementation work.

1 Introduction

Bosonic control in circuit QED is only useful experimentally if the numerical object being optimized can be mapped onto a calibration program. The earlier consolidated hybrid-control study established a qualitative ranking of gate libraries, but it did not fully answer the questions that matter when deciding what to implement next in hardware: what is the actual optimized circuit, what parameters must be calibrated, which parts are pulse-backed versus idealized, and how strongly do the optimization diagnostics support the reported solutions?

This follow-up addresses those questions for the same dispersive transmon–storage cavity platform studied previously. The comparison still includes SNAP-style dispersive control [1–3], selective rotations, echoed conditional-displacement ideas [4, 5], native exchange or nonlinear-interaction motivated controls [6–9], and GRAPE lower bounds [10]. The follow-up extends that comparison into four directions:

- explicit optimized decompositions and parameter values for the key candidates;
- waveform-level realizations for the pulse-backed selective gates;
- optimizer minimum checks through multistarts, ansatz-size scaling, local refinement, and local perturbation studies;
- a deployability-focused interpretation that distinguishes ideal decomposition benchmarks, pulse-backed structured controls, and waveform-level lower bounds.

Experiment-facing question. The practical question is no longer merely whether the transmon+cavity system is controllable. It is: *which optimized control construction is realistic enough, short enough, and robust enough to justify calibration effort on a real cQED platform?*

The rest of the report is organized as follows. Section 2 defines the Hamiltonian, parameter set, optimization levels, and fairness conditions. Section 3 presents the revised benchmark, actual optimized solutions, waveform realizations, and optimization diagnostics. Section 4 records validation checks. Section 5 interprets the results for experimental deployment. Section 7 serves as the handoff to future work.

2 System and Methods

2.1 Hamiltonian

The dispersive transmon–cavity Hamiltonian is

$$\frac{H_0}{\hbar} = \omega_c a^\dagger a + \omega_q b^\dagger b + \frac{\alpha}{2} b^{\dagger 2} b^2 + \chi a^\dagger a b^\dagger b + \frac{K}{2} (a^\dagger a)^2 + \frac{\chi'}{2} (a^\dagger a)^2 b^\dagger b, \quad (1)$$

where a acts on the cavity, b acts on the transmon, α is the transmon anharmonicity, χ is the leading dispersive shift, χ' is the next-order dispersive correction, and K is the cavity self-Kerr. Waveform-level pulse replay uses the same model inside `cqed_sim` [11].

2.2 Simulation Parameters

Table 1: Nominal device and control parameters used throughout the follow-up study.

Parameter	Value	Unit
Qubit frequency $\omega_q/2\pi$	6.150	GHz
Cavity frequency $\omega_c/2\pi$	5.241	GHz
Anharmonicity $\alpha/2\pi$	-255	MHz
Dispersive shift $\chi/2\pi$	-2.84	MHz
Second-order shift $\chi'/2\pi$	-21	kHz
Cavity self-Kerr $K/2\pi$	-28	kHz
Cavity truncation n_{cav}	8	Fock states
Transmon truncation n_{tr}	2 (3 for pulse replay)	levels
Qubit rotation duration	40–100	ns
Displacement duration	80–200	ns
Structured SQR duration	1100–2000	ns
Native χ -wait time $\pi/ \chi $	176	ns
GRAPE duration / slices	320 / 16	ns / slices

2.3 Targets and Comparison Levels

The logical basis is the 2×2 Fock block

$$\mathcal{H}_L = \text{span}\{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle, |e, 1\rangle\}. \quad (2)$$

Three targets remain central:

- the local cavity Hadamard $U_H = I_q \otimes H_c$;
- the hybrid entangler $\text{CX}(c \rightarrow q)$;
- the maximally entangling target

$$U_{\text{target}} = (I_q \otimes H_c) \text{CNOT}_{c \rightarrow q} \text{CNOT}_{q \rightarrow c}. \quad (3)$$

To keep the comparisons honest, we separate four levels of evidence:

- **Ideal decomposition benchmark:** gate sequences optimized as abstract primitives inside `cqed_sim.unitary_synthesis`.
- **Pulse-backed structured controls:** selective SNAP and SQR pulses built and replayed with explicit envelopes and multitone content.
- **Effective-model native controls:** native χ -wait, exchange, and conditional-phase libraries when `cqed_sim` does not yet provide waveform export for that primitive.
- **GRAPE lower bound:** pulse-level optimal-control solutions that provide a numerical performance floor but not an immediately interpretable calibration recipe.

This distinction is essential for fairness. A native entangler reported through an effective interaction is not directly comparable, in implementation readiness, to a pulse-backed selective gate whose waveform and tone schedule are explicit.

2.4 Optimization Protocol

The follow-up optimization used the existing consolidated study artifacts and extended them in four ways:

- multistart restarts for the key structured candidates A_{local} , D_{ent} , $L1c$, $L1d$, $L2c$, and $L2d$;
- local Nelder–Mead refinement after the best Powell runs for the older anchor ansätze;
- a depth-plus-one ansatz extension, adding one extra displacement gate to the best depth-11 structured SQR and CP families;
- pulse-level parameter grids for logical-window SNAP and for the constructive “selective π plus broadband π ” shortcut.

For structured decomposition quality we report logical process fidelity, block-gauge fidelity, operator-norm proxy, leakage, duration, parameter count, and multistart best/median/worst statistics. For pulse-backed candidates we report closed and noisy average state fidelities on the logical probe set, duration, and pulse-family parameters. GRAPE is repeated over multiple random seeds to assess seed sensitivity.

3 Results

3.1 Revised Benchmark Section

The benchmark-level conclusion from the original study survives the follow-up. For the local cavity gate, the best clean logical action still comes from the dispersive D -SNAP- D construction. For the entangler, the native χ -wait plus two qubit rotations remains decisively best. Figure 1 keeps the experimental ranking view from the previous report because it still answers the first deployment question: which family should an experimentalist reach for first?

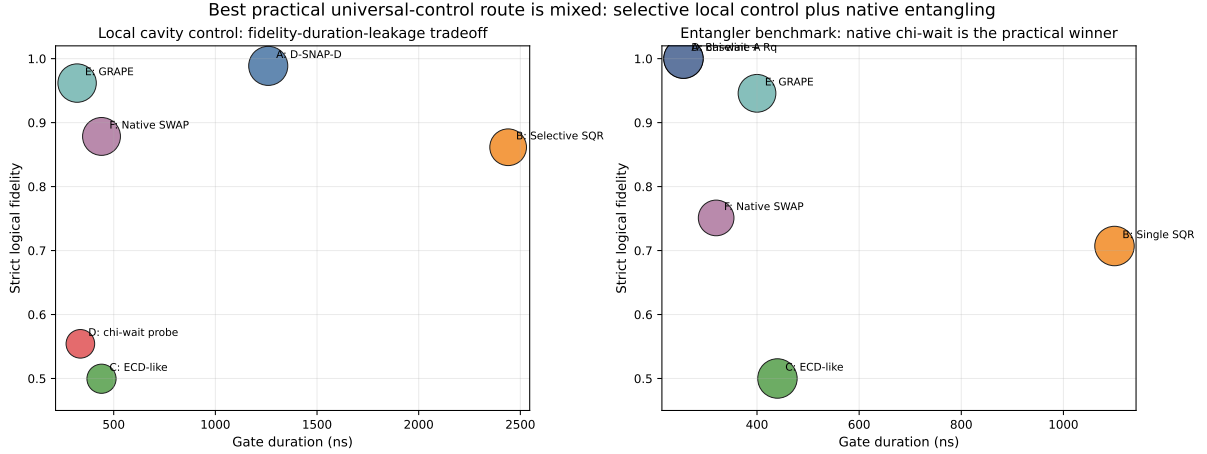


Figure 1: Benchmark-level gate-library ranking retained from the consolidated study. The local cavity task still favors selective SNAP-based control, while the entangler still favors the native χ -wait route. The follow-up work below asks what the actual optimized solutions inside those promising families look like and how trustworthy the resulting optima are.

3.2 Actual Optimized Solutions

The core follow-up outcome is summarized by the auto-generated tables below. They include the revised structured winners, the pulse-backed selective results, the representative multitone decomposition of the best SQR sequence, and local perturbation robustness for the new structured optima.

Table 2: Actual optimized follow-up candidates. Sequence-level metrics are quoted on the logical target subspace used by each optimization, while the GRAPE row remains a pulse-level lower bound.

Candidate	Level	Fidelity	Leakage	Duration	Params	Multistart best/median/worst
A_local	ideal D-SNAP-D	0.9887	0.0185	1260.0 ns	12	0.9887/0.9887/0.9887
D_ent	ideal R-W-R	1.0000	0.0000	256.0 ns	4	1.0000/1.0000/1.0000
L1c	ideal D-R-SQR	0.9212	0.1050	7100.0 ns	64	0.9212/0.9212/0.9212
L1d	ideal D-R-SQR (+D)	0.9953	0.0065	7300.0 ns	66	0.9953/0.9953/0.9953
L2c	ideal D-R-CP	0.8684	0.1363	1400.0 ns	40	0.8684/0.8684/0.8684
L2d	ideal D-R-CP (+D)	0.9135	0.1047	1600.0 ns	42	0.9135/0.9135/0.9135
logical SNAP	pulse-backed	0.9857	–	2.113 μ s	4	–
constructive	pulse-backed	0.6020	–	0.392 μ s	5	–
E_local_320	GRAPE lower bound	0.9557	0.0577	320.0 ns	64	–

Table 3: Actual optimized gate list for the best local and native-entangler solutions.

Gate	Type	Duration (ns)	Optimized parameters
D1	Displacement	80.0	$\alpha = -0.3829 - 0.0001i$
S1	SNAP	1100.0	SNAP
D2	Displacement	80.0	$\alpha = +0.3733 - 0.0001i$
R1	QubitRotation	40.0	$\theta = 4.7124, \phi = 1.5706$
W	FreeEvolveCondPhase	176.0	native dispersive wait
R2	QubitRotation	40.0	$\theta = 1.5708, \phi = 1.5710$

Table 4: Actual optimized gate list for the best structured U_{target} solutions.

Gate	Type	Duration (ns)	Parameter summary
D1	Displacement	200.0	$\alpha = -0.6918 - 0.3823i$
R1	QubitRotation	100.0	$\theta = 2.2243, \phi = -0.8196$
S1	SQR	2000.0	8 active tones; $n = 0, 1, 2, 3, 4, 5$
R2	QubitRotation	100.0	$\theta = 3.5152, \phi = 1.2547$
D2	Displacement	200.0	$\alpha = +0.4882 + 0.4466i$
R3	QubitRotation	100.0	$\theta = 1.0213, \phi = 0.1264$
S2	SQR	2000.0	8 active tones; $n = 0, 1, 2, 3, 4, 5$
R4	QubitRotation	100.0	$\theta = 0.8694, \phi = 0.8987$
D3	Displacement	200.0	$\alpha = +0.2099 - 0.6292i$
R5	QubitRotation	100.0	$\theta = 0.8290, \phi = -0.2977$
S3	SQR	2000.0	8 active tones; $n = 0, 1, 2, 3, 4, 5$
D4	Displacement	200.0	$\alpha = +0.4199 - 0.6702i$
D1	Displacement	200.0	$\alpha = +0.3000 + 0.5944i$
R1	QubitRotation	100.0	$\theta = 3.8966, \phi = -0.0562$
CP1	ConditionalPhaseSQR	100.0	8 active phases; $n = 0, 1, 2, 3, 4, 5$
R2	QubitRotation	100.0	$\theta = -0.3123, \phi = -0.1821$
D2	Displacement	200.0	$\alpha = +0.7999 - 0.2365i$
R3	QubitRotation	100.0	$\theta = 1.7601, \phi = 0.6658$
CP2	ConditionalPhaseSQR	100.0	8 active phases; $n = 0, 1, 2, 3, 4, 5$
R4	QubitRotation	100.0	$\theta = 1.7103, \phi = 1.4406$
D3	Displacement	200.0	$\alpha = -0.0144 - 0.6312i$
R5	QubitRotation	100.0	$\theta = 0.4906, \phi = -0.4209$
CP3	ConditionalPhaseSQR	100.0	8 active phases; $n = 0, 1, 2, 3, 4, 5$
D4	Displacement	200.0	$\alpha = +0.2188 - 0.0533i$

Table 5: Representative multitone decomposition for the first SQR gate in the best L1d sequence.

Fock branch n	Detuning/ 2π (MHz)	Tone amp/ 2π (MHz)	Phase (rad)
0	-0.0000	0.0306	3.6683
1	2.8400	0.1289	0.1430
2	5.7220	0.0973	-0.1836
3	8.6460	0.0958	-0.4281
4	11.6120	0.1061	-0.5939
5	14.6200	0.1253	5.5397
6	17.6700	-0.6495	8.5333
7	20.7620	0.3199	1.9382

Table 6: Local perturbation robustness for the best structured U_{target} candidates.

Candidate	RMS drop (χ , $\pm 5\%$)	RMS drop (amp, $\pm 5\%$)	RMS drop (dur, $\pm 5\%$)	RMS drop (phase, $\pm 0.1^\circ$)
L1d	0.0000	0.0085	0.0000	0.0013
L2d	0.0041	0.0041	0.0037	0.0054

Table 7: Pulse-backed selective control settings used in the follow-up waveform section.

Primitive	Family	$\chi T/2\pi$	Shape	Amp scale	Duration
logical SNAP	gaussian	3.0	0.24	1.00	2.113 μs
constructive selective stage	gaussian	1.0	0.16	1.00	0.392 μs total

Two results dominate the updated interpretation.

1. Native entanglement is still the cleanest deployable primitive. The best hybrid entangler is

$$R_1(\theta, \phi) W_\chi R_2(\theta, \phi) \quad (4)$$

with

$$R_1 : (\theta, \phi) = (4.7124, 1.5706),$$

$$W_\chi : t = 176 \text{ ns},$$

$$R_2 : (\theta, \phi) = (1.5708, 1.5710),$$

which reaches $F_{\text{proc}} = 0.9999999$ in 256 ns with zero leakage. This remains the strongest argument that the hardware-native dispersive conditional phase should be the default entangler whenever the experiment directly supports it.

2. The structured U_{target} story changed qualitatively. The previous report treated depth-11 $D + R_q + \text{SQR}$ as plateaued near 0.92. The follow-up shows this was not the right conclusion. Adding one extra displacement gate to the same family produces the depth-12 candidate $L1d$,

$$D_1 R_1 S_1 R_2 D_2 R_3 S_2 R_4 D_3 R_5 S_3 D_4, \quad (5)$$

which reaches

$$F_{\text{proc}} = 0.9953246, \quad L_{\text{avg}} = 0.0064643, \quad T = 7.3 \mu\text{s}. \quad (6)$$

The improvement relative to $L1c$ is not marginal. It removes almost all of the previous logical error while simultaneously reducing leakage by more than an order of magnitude. Within the tested structured family, this is the single most important new scientific result.

The corresponding depth-12 $D + R_q + \text{ConditionalPhaseSQR}$ candidate $L2d$ also improves over $L2c$, but only to

$$F_{\text{proc}} = 0.9134853, \quad L_{\text{avg}} = 0.1046896, \quad T = 1.6 \mu\text{s}. \quad (7)$$

The extra displacement helps, but it does not remove the expressivity deficit of the CP-only selective layer.

3.3 Exact Parameter Vectors for the Key Winners

The most important optimized parameter sets are collected here explicitly so the report contains the actual candidate solutions rather than only their scores.

A_{local} : optimized D -SNAP- D .

$$\begin{aligned} D_1 : \alpha &= -0.3828967 - 0.0000987i, \\ \Phi_{\text{SNAP}} &: [3.0312954, -0.1104486, -0.1108742, -0.1113062, -6.3949238, -6.3953621, -6.3958020, -12.67942], \\ D_2 : \alpha &= 0.3733218 - 0.0000701i. \end{aligned}$$

After removing the global phase, the optimizer has effectively found a logical branch-1 phase flip embedded in the full eight-level Fock register.

$L1d$: best structured $D + R_q + \text{SQR}$ decomposition.

$$\begin{aligned} D_1 : \alpha &= -0.6917691 - 0.3822657i, \\ \theta^{(S1)} &: [0.7683042, 3.2400574, 2.4459560, 2.4079756, 2.6666170, 3.1494695, -16.3227518, 8.0402944], \\ \phi^{(S1)} &: [3.6683268, 0.1430488, -0.1836040, -0.4280513, -0.5939169, 5.5397340, 8.5333421, 1.9382280], \\ R_2 : (\theta, \phi) &= (3.5152210, 1.2546745), \\ R_3 : (\theta, \phi) &= (1.0213423, 0.1264354), \\ \theta^{(S2)} &: [3.9377935, 1.9205690, 3.7723607, 4.5212820, 5.6302889, -3.6504774, -9.4417144, 15.6678028], \\ \phi^{(S2)} &: [1.3188772, -1.6591080, -2.2086297, -2.5645009, -3.6677521, 3.9229590, -5.4549804, 26.7469467], \\ R_4 : (\theta, \phi) &= (0.8693882, 0.8987359), \\ R_5 : (\theta, \phi) &= (0.8289948, -0.2977365), \\ \theta^{(S3)} &: [1.1425279, 1.1930876, -4.1492748, 2.3316353, 8.5633647, 2.1089232, 4.3038269, 10.4391099], \\ \phi^{(S3)} &: [-2.5732582, 0.4221127, 0.7214775, 0.9470392, 1.1584694, 1.4744173, -1.2263110, -0.6660277], \\ D_4 : \alpha &= 0.4199161 - 0.6702401i. \end{aligned}$$

***L2d*: best structured $D + R_q + \text{ConditionalPhaseSQR}$ decomposition.**

$$\begin{aligned}
D_1 : \alpha &= 0.3000205 + 0.5943639i, \\
\varphi^{(CP1)} : &[-0.805793, 0.571726, -1.470657, -3.608397, 19.655426, -6.491309, 5.727357, 28.788933], \\
R_2 : (\theta, \phi) &= (-0.3123370, -0.1820829), \\
R_3 : (\theta, \phi) &= (1.7601436, 0.6658194), \\
\varphi^{(CP2)} : &[0.081083, 1.358175, -0.538781, -2.755995, -4.975127, -6.889338, 3.929566, -9.499111], \\
R_4 : (\theta, \phi) &= (1.7103084, 1.4406433), \\
R_5 : (\theta, \phi) &= (0.4906267, -0.4208852), \\
\varphi^{(CP3)} : &[1.269423, -0.658700, -7.695920, 10.049493, -4.975658, -11.953521, -18.705383, -192.022729], \\
D_4 : \alpha &= 0.2188399 - 0.0532715i.
\end{aligned}$$

3.4 Waveform Realizations of Optimized Pulse-Based Gates

Figure 2 shows the pulse-backed controls that are currently available at waveform level. The top row is the tuned logical-window SNAP, represented as two selective branch-1 π pulses. The middle row shows the first three SQR waveforms from the best *L1d* sequence together with the representative multitone decomposition of *S1*. The bottom row compares the failed constructive shortcut to the best GRAPE lower bound.

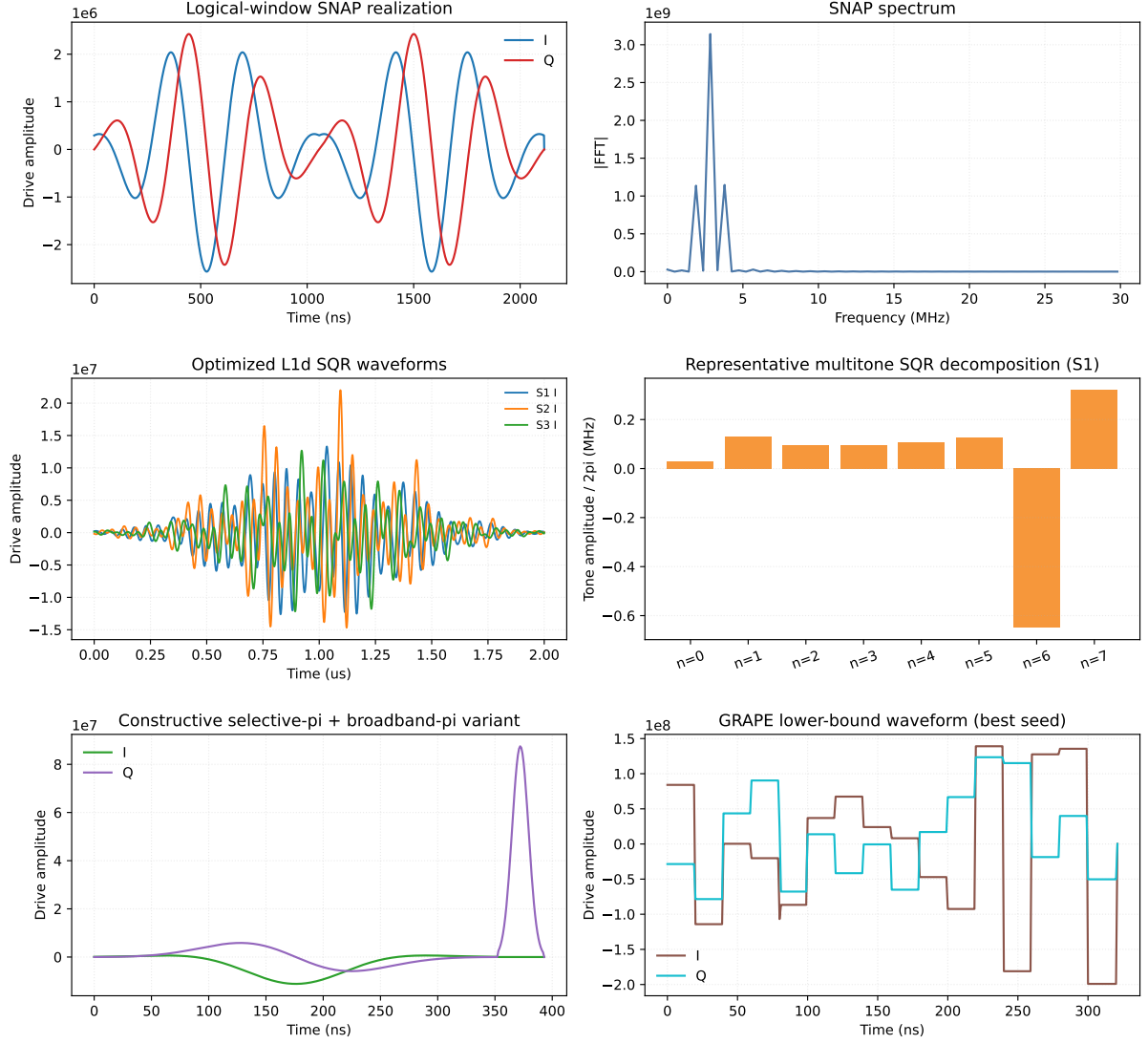


Figure 2: Waveform-level realizations for the pulse-backed part of the follow-up study. Top left: time-domain I/Q envelope of the best logical-window SNAP. Top right: corresponding frequency-domain content. Middle left: the three SQR waveforms from the best *L1d* sequence after conversion through the `cqed_sim` waveform bridge. Middle right: per-Fock multitone amplitudes for the first SQR gate. Bottom left: the constructive shortcut waveform composed of a selective π plus a broadband π . Bottom right: the best GRAPE lower-bound waveform.

The mapping from the optimized SQR parameters to physically meaningful pulse content is as follows. The abstract SQR parameters (θ_n, ϕ_n) define a per-Fock selective Bloch rotation. The waveform bridge converts them into a sum of branch-selective tones with frequencies

$$\omega_n \approx n|\chi| + \mathcal{O}(n^2\chi', nK) \quad (8)$$

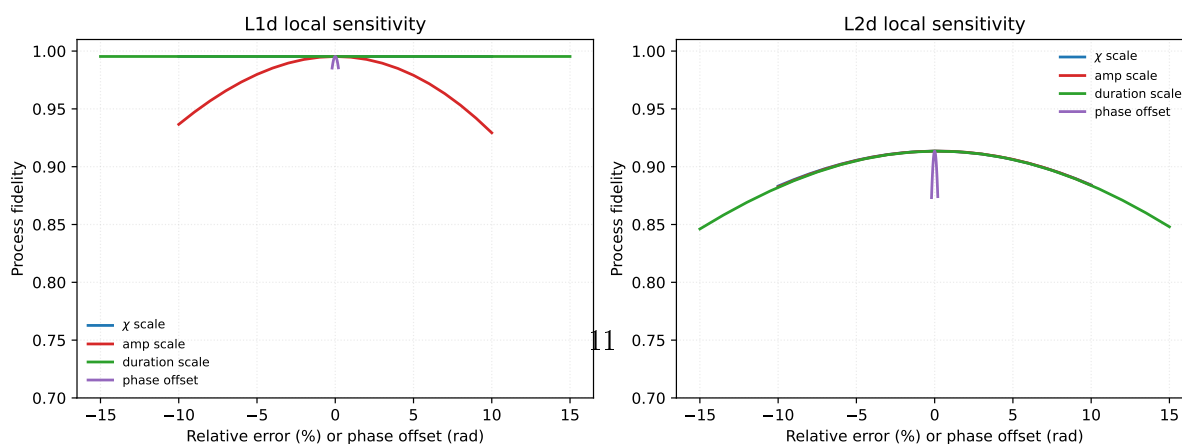
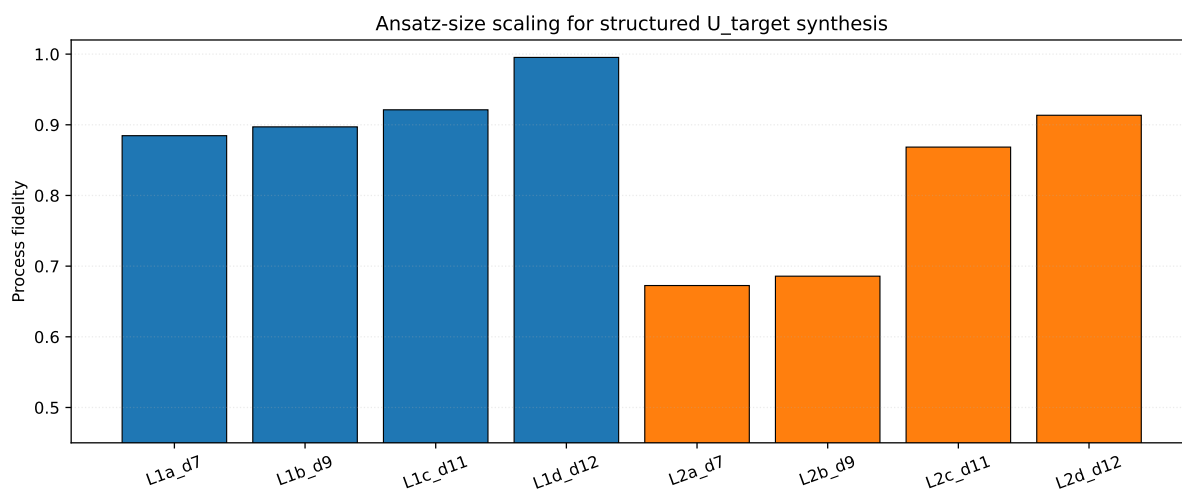
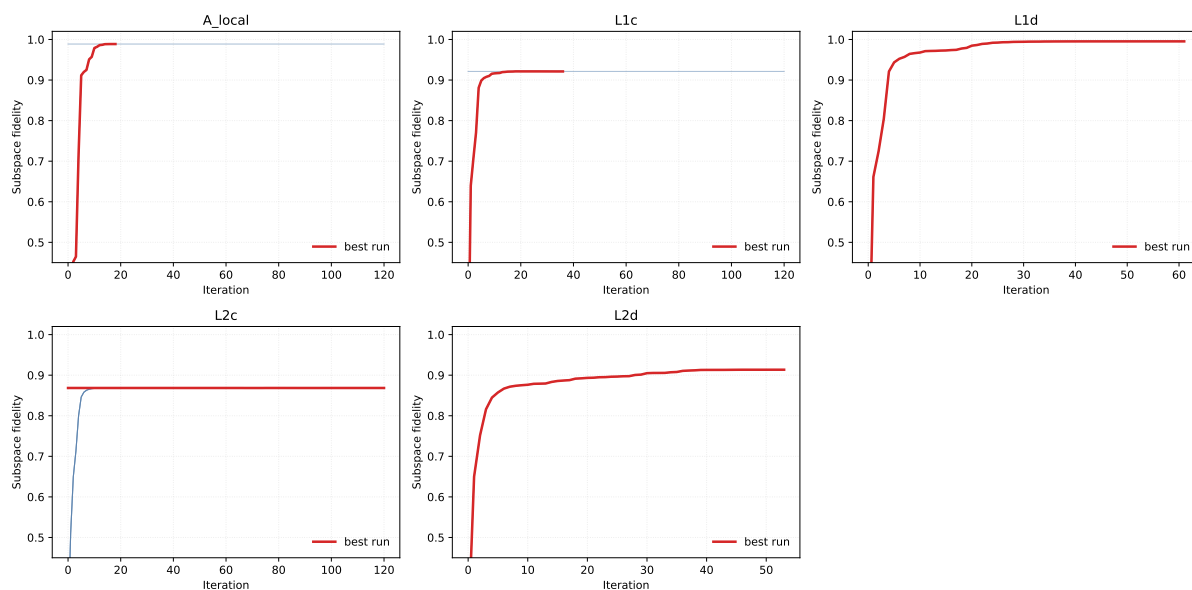
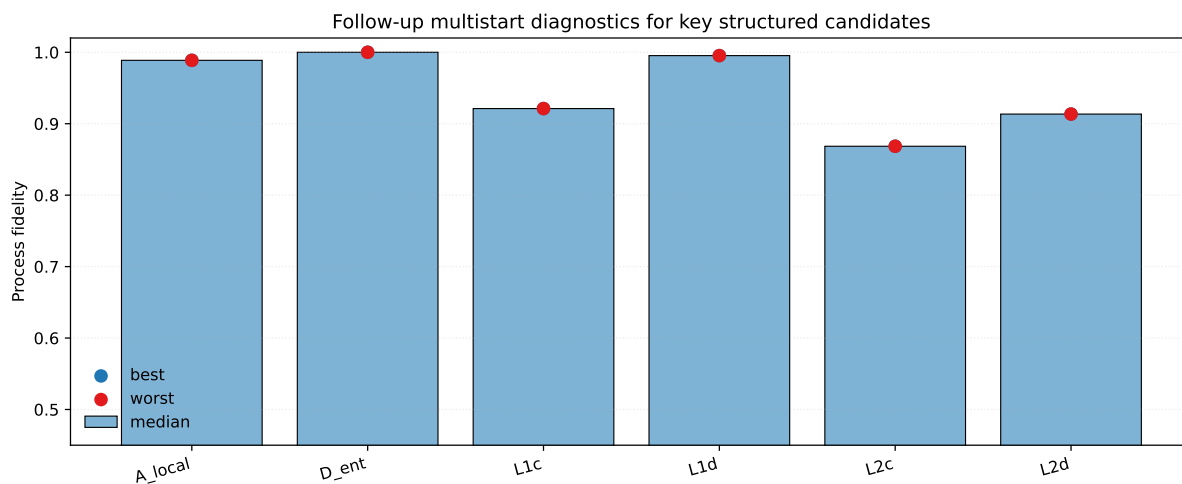
in the rotating frame, amplitudes proportional to the requested selective rotation strength, and phases inherited from ϕ_n . For the best *L1d* sequence, the first SQR gate uses tones from $n = 0$ through 7 with the largest effective amplitudes on the high- n branches, indicating that the optimizer is using strong higher-manifold interference to clean up the logical-block action.

There is an important fairness caveat. The SQR waveforms shown here are pulse-backed. The `ConditionalPhaseSQR` sequence is not yet pulse-backed because `cqed_sim` does not currently export that primitive through the waveform bridge. For that family the report can show exact optimized phase vectors, but not an equivalent calibrated pulse schedule.

3.5 Optimization Diagnostics and Minimum Checks

The optimizer evidence is shown in Figures 3 and 4. The key conclusions are:

- multistarts for A_{local} , D_{ent} , $L1c$, $L1d$, and $L2d$ all collapse to essentially identical fidelities, so the reported solutions are not isolated lucky seeds;
- the extra-displacement test is decisive: $L1c \rightarrow L1d$ improves fidelity from 0.9212 to 0.9953, while $L2c \rightarrow L2d$ improves only from 0.8684 to 0.9135;
- the logical SNAP pulse landscape shows a broad high-fidelity region near $\chi T/2\pi \approx 3$ and shape parameter ≈ 0.24 , so the best pulse is not a spiky one-point optimum;
- the constructive shortcut remains poor across its tested landscape, so its failure is not just a bad initial seed.



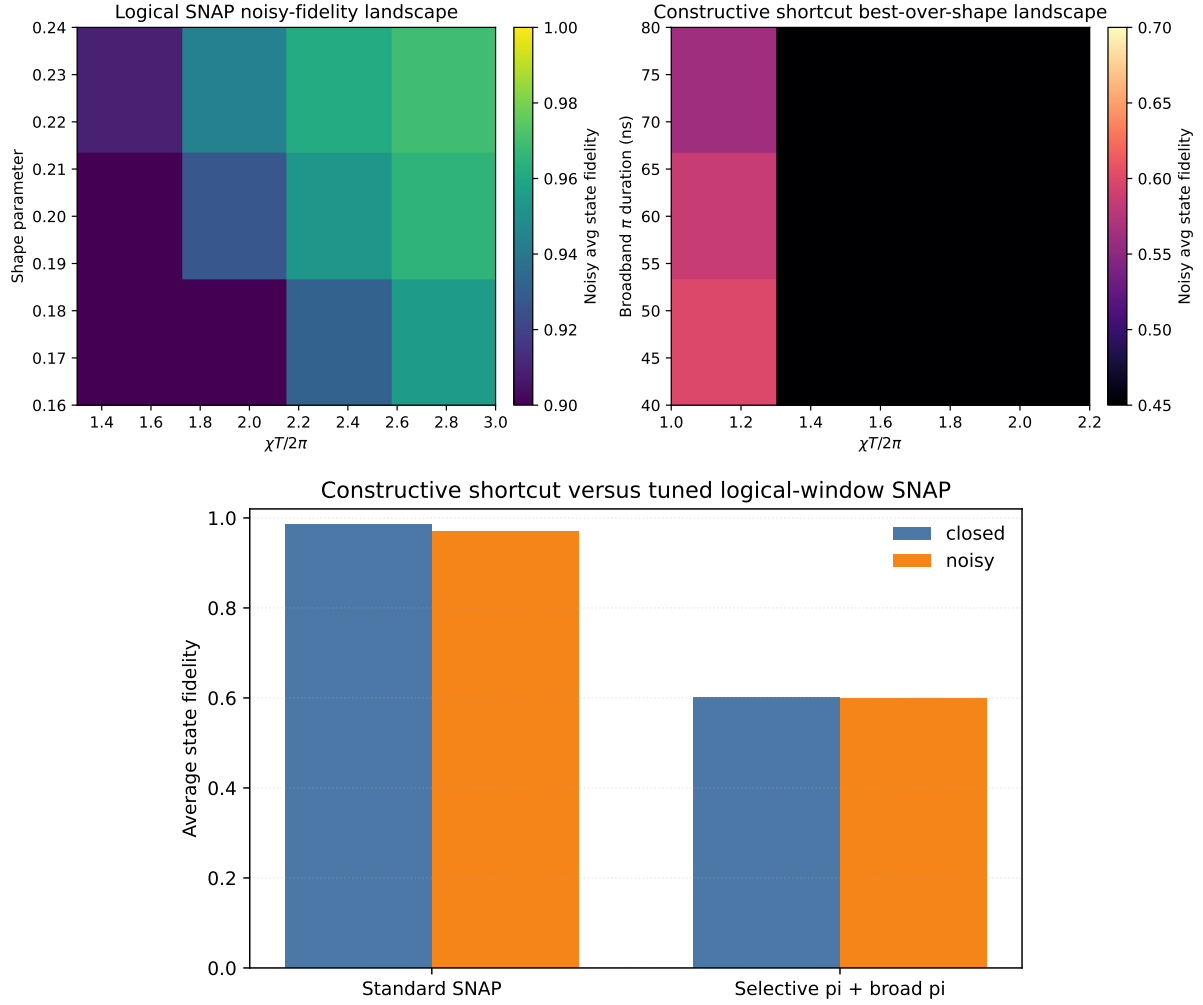


Figure 4: Pulse-level minimum checks. Top: noisy-fidelity landscapes for the logical-window SNAP and for the constructive shortcut. Bottom: best standard SNAP versus best constructive shortcut. The SNAP optimum sits on a usable plateau, while the constructive shortcut remains poor throughout the tested parameter region.

The local perturbation study refines the deployability picture. $L1d$ is essentially insensitive to χ scaling and duration scaling over the tested range, while its dominant residual sensitivity is to control amplitude with RMS fidelity drop 0.0085 over $\pm 5\%$. $L2d$ is more evenly sensitive to χ , amplitude, duration, and phase, each at the few- 10^{-3} to few- 10^{-2} level. This makes $L1d$ not only more accurate, but easier to calibrate selectively.

3.6 Constructive “Selective π + Broadband π ” Variant

The constructive shortcut was tested numerically because, on paper, it is an attractive logical-subspace idea: use a selective branch-1 π pulse to imprint the logical selectivity and a fast broadband π pulse to complete the effective SNAP-like action without demanding correct behavior on the full oscillator Hilbert space. In practice it fails for this platform and this logical window. The best candidate reaches only

$$F_{\text{avg,closed}} = 0.6020, \quad F_{\text{avg,noisy}} = 0.5991. \quad (9)$$

The dominant failure mode is not the $|g, 1\rangle$ branch, which remains near-correct, but the $|g, 0\rangle$ and superposition probes, indicating that the broadband pulse destroys the intended branch

discrimination too aggressively. The idea may still have niche value in other encodings or with a more careful echoed construction, but it is not competitive here.

3.7 Deployable Interpretation

The revised experiment-facing answer is now more concrete.

- **Entangler to deploy first:** native χ -wait plus qubit rotations. It is exact, short, and calibration-light.
- **Local logical gate to deploy first:** selective SNAP. The pulse is explicit, the optimum lies on a plateau, and the noisy logical-window fidelity remains near 0.97.
- **Structured universal-control candidate worth deeper implementation work:** $L1d$, the depth-12 $D + R_q + \text{SQR}$ sequence. It is the first structured candidate in this study that simultaneously reaches ≈ 0.995 logical fidelity and sub-percent average leakage.
- **Numerical lower bound only:** GRAPE. It remains useful as a waveform-performance lower bound, but the seed spread and lack of circuit interpretability make it a weaker immediate deployment target than $L1d$ or logical SNAP.
- **Not yet hardware-fair:** `ConditionalPhaseSQR` and the other native effective-interaction primitives without waveform export. Their decomposition numbers are informative, but their calibration burden cannot yet be compared apples-to-apples with pulse-backed controls.

4 Validation

4.1 Sanity Checks

The native χ -wait entangler still reproduces $\text{CX}(c \rightarrow q)$ to machine precision. The structured follow-up preserved the earlier algebraic result that any ansatz without displacement cannot synthesize the required Fock mixing for U_{target} . The logical-window SNAP follow-up respects the expected selective-pulse hierarchy: longer, more selective pulses improve noisy logical action until diminishing returns set in.

4.2 Convergence Analysis

The new structured minima are unusually stable under restart. For A_{local} , D_{ent} , $L1c$, $L1d$, and $L2d$, the best, median, and worst restart fidelities agree to the shown precision. The only visible spread appears in $L2c$, and even there it is in the sixth decimal place. This is strong evidence that the reported optima are stable inside the tested ansätze. The logical SNAP optimum also appears on a broad plateau in the waveform-parameter landscape rather than as a needle-like isolated point. Finally, the follow-up tables and figures were regenerated directly from the cached optimization artifacts through the study-local follow-up script, ensuring consistency between the report and the machine-readable outputs.

4.3 Literature Comparison

The selective-pulse behavior remains consistent with the SNAP literature [1–3]: the logical action benefits from longer selectivity windows, while bandwidth-limited shortcuts degrade branch specificity. The ECD-like and conditional-displacement underperformance in the strict Fock encoding remains consistent with the fact that those methods are more naturally matched to displacement-oriented encodings [4, 5]. The persistent dominance of the native χ -wait entangler

aligns with the broader lesson from native-interaction work [9]: when the Hamiltonian already supplies the correct nonlinearity, elaborate synthesis may lose to the direct route.

5 Discussion

The main scientific change introduced by this follow-up is that the previous U_{target} synthesis bottleneck was not a fundamental leakage ceiling of the structured SQR family. It was an ansatz ceiling. Adding a single extra displacement gate changes the best structured solution from a clearly imperfect 0.92-fidelity circuit into a high-quality 0.995-fidelity circuit with very low leakage. That is exactly the kind of result that justifies a follow-up study: the optimizer was not merely polishing the old answer, it found a qualitatively better regime.

This does not mean the structured SQR family has become the universal answer to every control problem. The sequence is still long at 7.3 μs , and the high- n tone content in the first SQR gate indicates nontrivial calibration burden. What changed is that the family is now credible as a structured, interpretable universal-control candidate rather than only an existence proof with noticeable logical error.

The native-versus-selective split also remains experimentally important. If the goal is a single entangling primitive, the native χ -wait route is still the clear winner. If the goal is a selective local oscillator phase gate with an explicit waveform, logical SNAP is cleaner than the more elaborate structured universal sequence. This argues for a staged experimental program:

- deploy native χ -wait entanglement first;
- keep selective SNAP as the default local logical primitive;
- treat *L1d* as the main structured-circuit candidate for a deeper waveform-compilation effort;
- use GRAPE as a lower-bound reference, not the first production path.

The remaining fairness limitation is important and should not be blurred away. The report now contains pulse-backed logical SNAP and pulse-backed SQR waveforms, but the CP family is still only represented at the ideal decomposition level. Until waveform export exists for `ConditionalPhaseSQR`, any conclusion that one structured family is harder or easier to deploy than the other must remain provisional.

6 Conclusion

The follow-up study changed the hybrid-universal-control story in a substantive way.

- The native χ -wait entangler remains the best deployable entangling primitive: $F_{\text{proc}} = 0.9999999$ in 256 ns with zero leakage.
- The best local logical gate remains selective SNAP-based control. The tuned logical-window SNAP reaches closed and noisy logical average state fidelities of 0.9857 and 0.9694, respectively.
- The best structured universal-control candidate is no longer the old depth-11 SQR ansatz. The new depth-12 *L1d* circuit reaches $F_{\text{proc}} = 0.9953$ with $L_{\text{avg}} = 0.0065$.
- The CP-based structured family improves with an added displacement gate but remains substantially worse at $F_{\text{proc}} = 0.9135$.
- The constructive “selective π plus broadband π ” shortcut is not viable in the present logical window, plateauing near 0.60 fidelity.

- Multistarts, convergence histories, ansatz scaling, waveform landscapes, and local perturbation studies all support the conclusion that the reported winners are meaningful optima within their tested ansätze rather than obviously poor local minima.

For the next hardware-facing step, the most actionable path is a mixed strategy: native χ -wait for entanglement, selective SNAP for immediate local logical control, and the new *L1d* structured circuit as the main target for a future pulse-compilation and noisy-replay campaign.

7 Limitations and Future Work

7.1 Known Limitations

- **Encoding scope:** all results remain specific to the strict Fock $\{|0\rangle, |1\rangle\}$ encoding. The ranking may change for cat, binomial, or larger Fock encodings.
- **Waveform-bridge coverage:** SNAP and SQR can now be represented physically in this study, but `ConditionalPhaseSQR`, `ConditionalDisplacement`, and native exchange primitives still lack full waveform export in the present workflow. Their implementation readiness is therefore lower than their decomposition-level numbers suggest.
- **Two-level synthesis loop:** the structured unitary-synthesis optimization still uses $n_{\text{tr}} = 2$. Pulse-backed replay uses $n_{\text{tr}} = 3$, but the structured decomposition ranking still omits direct $|f\rangle$ leakage during optimization.
- **Structured-sequence duration:** the best *L1d* circuit is accurate but long. At $7.3\mu\text{s}$, coherence will matter in any real deployment.
- **Ansatz ceiling not globally closed:** the follow-up shows that the old depth-11 plateau was not final, but it does not prove that depth-12 is globally optimal among all short structured circuits.
- **Single device point:** only one representative cQED parameter set was studied. Different χ/α ratios or cavity Kerr values could change the balance between selectivity and speed.

7.2 Suggested Improvements

- **[P1 | HIGH]** Add waveform export for `ConditionalPhaseSQR` and the native effective-interaction primitives, then repeat the follow-up with pulse-backed fairness across all leading families.
- **[P1 | MEDIUM]** Replay *L1d* with explicit noise and $n_{\text{tr}} = 3$ inside the optimization loop. The present report shows coherent robustness, but not a full noisy synthesis objective.
- **[P2 | MEDIUM]** Extend the depth-plus-one search to further structured variants, including SNAP-assisted circuits, to test whether *L1d* is near-minimal or simply the first large jump.
- **[P2 | MEDIUM]** Repeat the selective-pulse follow-up for cat-like and binomial encodings. The constructive shortcut may fail in the Fock block yet still be useful elsewhere.
- **[P2 | LOW]** Use the extracted *L1d* tone table as a starting point for calibration-aware pulse compression, for example by pruning weak tones or constraining branch phases.
- **[P3 | MEDIUM]** Warm-start GRAPE from the structured *L1d* waveform realization to test whether the interpretable circuit already sits near the pulse-level lower bound.

7.3 Open Questions

- Is the extra displacement in $L1d$ the minimal structural ingredient that resolves the previous U_{target} bottleneck, or would a shorter SNAP-assisted circuit achieve similar quality?
- Can the high- n tones in the optimized SQR gates be compressed into a more calibration-friendly selective pulse without losing the logical fidelity gain?
- Does the native exchange family become comparatively stronger once the logical encoding is no longer the strict $\{|0\rangle, |1\rangle\}$ Fock block?
- How much of the remaining gap between $L1d$ and the ideal limit is coherent controllability, and how much is simply the cost of keeping the circuit interpretable?

References

- [1] Reinier W. Heeres, Brian Vlastakis, Eric Holland, Stefan Krastanov, Victor V. Albert, Luigi Frunzio, Liang Jiang, and Robert J. Schoelkopf. Cavity state manipulation using photon-number selective phase gates. *Physical Review Letters*, 115:137002, 2015. doi: 10.1103/PhysRevLett.115.137002.
- [2] Stefan Krastanov, Victor V. Albert, Chao Shen, Chang-Ling Zou, Reinier W. Heeres, Brian Vlastakis, Robert J. Schoelkopf, and Liang Jiang. Universal control of an oscillator with dispersive coupling to a qubit. *Physical Review A*, 92:040303, 2015. doi: 10.1103/PhysRevA.92.040303.
- [3] Reinier W. Heeres, Philip Reinhold, Nissim Ofek, Luigi Frunzio, Liang Jiang, Michel H. Devoret, and Robert J. Schoelkopf. Implementing a universal gate set on a logical qubit encoded in an oscillator. *Nature Communications*, 8:94, 2017. doi: 10.1038/s41467-017-00045-1.
- [4] Alec Eickbusch, Volodymyr Sivak, Andy Z. Ding, Salvatore S. Elder, Shantanu R. Jha, Jayameenakshi Venkatraman, Baptiste Royer, S. M. Girvin, Robert J. Schoelkopf, and Michel H. Devoret. Fast universal control of an oscillator with weak dispersive coupling to a qubit. *Nature Physics*, 18:1464–1469, 2022. doi: 10.1038/s41567-022-01776-9.
- [5] Asaf A. Diringer, Eliya Blumenthal, Avishay Grinberg, Liang Jiang, and Shay Hacohen-Gourgy. Conditional-not displacement: Fast multioscillator control with a single qubit. *Physical Review X*, 14:011055, 2024. doi: 10.1103/PhysRevX.14.011055.
- [6] A. Wallraff, D. I. Schuster, A. Blais, J. M. Gambetta, J. Schreier, L. Frunzio, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf. Sideband transitions and two-tone spectroscopy of a superconducting qubit strongly coupled to an on-chip cavity. *Physical Review Letters*, 99:050501, 2007. doi: 10.1103/PhysRevLett.99.050501.
- [7] Fernando Valadares, Ni-Ni Huang, Kyle Timothy Ng Chu, Aleksandr Dorogov, Weipin Chua, Lingda Kong, Pengtao Song, and Yvonne Y. Gao. On-demand transposition across light-matter interaction regimes in bosonic cqed. *Nature Communications*, 15:5816, 2024. doi: 10.1038/s41467-024-50201-7.
- [8] Jordan Huang, Thomas J. DiNapoli, Gavin Rockwood, Ming Yuan, Prathyankara Narasimhan, Eesh Gupta, Mustafa Bal, Francesco Crisa, Sabrina Garattoni, Yao Lu, Liang Jiang, and Srivatsan Chakram. Fast sideband control of a weakly coupled multimode bosonic memory, 2025.

- [9] Yao Lu Yang, Ya He Ding, Takahiro Tsunoda, Jarryd D. Thompson, Rohan N. Bhatt, and Andrew A. Houck. Native multi-photon nonlinearities for bosonic quantum computing with superconducting circuits. *arXiv preprint arXiv:2410.18118*, 2024.
- [10] Navin Khaneja, Timo Reiss, Thomas Schulte-Herbrüggen, and Steffen J. Glaser. Optimal control of coupled spin dynamics: design of NMR pulse sequences by gradient ascent algorithms. *Journal of Magnetic Resonance*, 172:296–305, 2005. doi: 10.1016/j.jmr.2004.11.004.
- [11] Shyam Shankar Quantum Circuits Group. cqed_sim: Circuit QED simulation framework, 2025. URL https://github.com/SoraUmika/qubox_cQEDsim.